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HOT TOPIC

Esophageal Intelligence: Implementing Artificial Intelligence Into the Diagnostics of Esophageal Motility and Impedance pH Monitoring

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ABSTRACT

Esophageal motility disorders (EMDs) encompass a range of functional abnormalities, including achalasia, ineffective esophageal motility (IEM), esophagogastric junction outflow obstruction (EGJOO), and distal esophageal spasm (DES). Diagnostic modalities like high-resolution esophageal manometry (HREM), Functional Lumen Imaging Probe (FLIP), and impedance analysis are invaluable but often limited by interpretive variability and the need for expert analysis. Artificial intelligence (AI) has emerged as a transformative tool in addressing these challenges. This manuscript explores the integration of AI in EMD diagnostics, showcasing its ability to enhance diagnostic accuracy, optimize workflows, and standardize interpretation across centers. Advanced algorithms, including convolutional neural networks (CNNs) and machine learning (ML) models, achieve high accuracy in automating classifications, subtyping disorders like achalasia, and improving diagnostic consistency. Furthermore, AI's predictive capabilities extend to treatment outcome modeling, enabling personalized care strategies and longitudinal tracking. AI also offers significant potential in medical education by reducing learning curves and standardizing esophageal motility interpretation training. These advancements collectively emphasize the role of AI in revolutionizing EMD diagnosis, treatment, and training, promising improved patient outcomes and broader clinical utility.

1 | Introduction

Esophageal motility disorders (EMDs) encompass a range of functional abnormalities, including achalasia, ineffective esophageal motility (IEM), esophagogastric junction outflow obstruction (EGJOO), distal esophageal spasm (DES), and hypercontractile esophagus [1, 2]. These conditions, characterized by disrupted esophageal contractions and/or impaired lower esophageal sphincter (LES) relaxation, significantly impact patient quality of life [3, 4]. High-resolution esophageal manometry (HREM) is the main diagnostic tool, complemented by technologies like Functional Lumen Imaging Probe (FLIP) panometry and impedance analysis [5–10]. Despite their utility, these methods demand expert interpretation and are prone to inter- and intra-observer variability [11]. Artificial intelligence (AI) has emerged as a potential transformative force in this field, streamlining workflows, improving diagnostic accuracy, saving time, and reducing variability [12–15]. This review will examine the role of AI in esophageal motility, with discussions on machine learning (ML)–a subset of AI that enables systems to learn from data, refine decision-making, and improve autonomously over time. While ML is not the sole component of AI, its ability

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Summary

- Artificial intelligence (AI) enhances the diagnosis of esophageal motility disorders by automating data analysis from tools like high-resolution esophageal manometry and functional lumen imaging probe (FLIP), achieving high accuracy, reducing inter- and intra-observer variability, and optimizing workflows.
- AI models, such as those integrating biomechanical data, predict treatment outcomes, refine therapy strategies, and enable patient-specific interventions, improving long-term management of disorders like achalasia.
- AI may facilitate medical training by standardizing diagnostic interpretations, offering virtual mentorship, and reducing the learning curve for complex esophageal motility tests, promoting consistent and accurate results across practitioners and centers.

to enhance analytical precision and adaptability makes it a valuable aspect of AI's application in this field.

2 | AI in Esophageal Diagnostic Tests

AI has profoundly impacted the interpretation of diagnostic tests for EMDs, including HREM, FLIP panometry, and impedance analysis [6, 15–19]. These tests generate intricate datasets that require expert evaluation to identify motility patterns and diagnose motility disorders. However, human interpretation is prone to variability, especially in borderline cases [11]. AI algorithms, particularly deep learning (DL) and ML, address these limitations by automating data processing and enhancing diagnostic precision [14, 15].

High-Resolution Esophageal Manometry (HREM): HREM remains the gold standard for evaluating EMDs [20]. Surdea-Blaga et al. developed an AI-based platform using convolutional neural networks (CNNs) to automate the classification of motility disorders, achieving an accuracy of 86% in categorizing 10 diagnostic subtypes, including achalasia subtypes and fragmented peristalsis [14]. This tool reduced interpretation time by 50%, addressing the bottleneck of manual analysis [14].

Functional Lumen Imaging Probe (FLIP) Panometry is increasingly used to assess esophageal distensibility and impairment of secondary motility. Kou et al. integrated CNNs into FLIP data analysis, achieving 89% accuracy in diagnosing achalasia [15]. Of the patients diagnosed with achalasia with HREM, zero patients were predicted as not affected and 93% were diagnosed with achalasia in the AI model [15]. These AI models provided consistent results across varying levels of operator expertise, enhancing the utility of FLIP in clinical practice.

Impedance Analysis: Impedance measurements, often coupled with HREM, assess bolus transit and esophageal clearance. Zifan et al. evaluated the performance of the three AI algorithms (LogitBoost, SVM, and gradient boosting) against traditional methods (the 5-mL bolus in the supine position to assess esophageal motility) by using receiver operating characteristic (ROC) curves to assess their ability to classify patients with functional dysphagia and normal subjects. The AI algorithms, particularly the SVM model, outperformed the non-AI methods by achieving higher accuracy, especially in classifying based on distension features, when compared to the standard diagnostic protocol (the 5-mL bolus in the supine position) [21]. Furthermore, By developing an advanced decision tree algorithm, AI can accurately identify, censor, and remove esophageal events to calculate artifact-free baseline impedance (BI) values. Notably, the AI-derived upright-to-recumbent BI ratio (U:R AIBI ratio) offers a novel metric that correlates with GERD treatment outcomes, particularly in cases with borderline acid exposure time [19].

Despite these capabilities, the role of AI models has yet to be validated in the context of atypical clinical presentations and in patients with multiple comorbidities. Future studies are needed to explore these abilities and implementations.

3 | Automated Classification of Esophageal Motility Disorders

The Chicago Classification system classifies the different EMDs according to specific metrics like distal latency, contractile deceleration points, and integrated relaxation pressure (IRP). However, their manual identification is prone to different degrees of variability, often dependent on the operators' experience [22, 23]. AI has addressed this limitation by automating the classification process.

CNN Models for Chicago Classification: Popa et al. developed a CNN model to automate the classification of esophageal motility disorders based on the Chicago Classification [18]. They utilized a CNN model to analyze HREM data. The results of their study showed that the CNN model achieved high accuracy in two key tasks: Probe positioning detection and IRP classification. The model successfully identified the position of the esophageal probe, with a high level of accuracy [18]. It also effectively classified IRP, which is a key metric in the Chicago Classification for diagnosing disorders like achalasia [18]. The CNN model achieved classification accuracies above 90% for both tasks, indicating that deep learning can be highly effective in automating the Chicago Classification and improving diagnostic processes for esophageal motility disorders. This could help reduce human intervention and enhance diagnostic efficiency [18].

Advanced Metrics Integration: Traditional Chicago Classification relies on pressure metrics alone, but AI systems have incorporated secondary parameters like impedance and contraction vigor. Kou et al. demonstrated that machine learning algorithms integrating these metrics improved classification accuracy for conditions like EGJOO and fragmented peristalsis, achieving higher diagnostic precision and better differentiation between motility disorders compared to using pressure metrics alone [15].

Improved Consistency Across Centers: Multicenter studies have validated AI's generalizability in applying the Chicago Classification. Surdea-Blaga et al. demonstrated that AI tools,

such as deep learning models, showed consistent performance across diverse populations, achieving an accuracy of 86% and high F1-scores in classifying esophageal motility disorders. Their study emphasized the potential of AI to act as standardized diagnostic aids, improving consistency and reducing inter-center variability in diagnosis [14].

3.1 | Achalasia Diagnosis and Subtyping

Achalasia diagnosis requires precise subtyping (Type I, II, III) for effective treatment planning [24, 25]. AI systems have excelled in identifying these subtypes, overcoming challenges posed by overlapping features and subtle variations [13].

Improved Subtyping with ML: Carlson et al. applied machine learning algorithms to FLIP panometry data, achieving 71% accuracy in differentiating type III achalasia from non-spastic achalasia in the training cohort [26]. By incorporating additional metrics like bolus transit and DCI, the models enhanced sensitivity and specificity. The study also identified key features, such as bolus transit time and DCI, as critical for accurate classification, demonstrating the potential of machine learning to improve the subtyping and diagnostic precision of achalasia [26].

Treatment prediction: AI's role extends beyond diagnosis to predicting treatment outcomes. Halder et al. developed a novel framework, the mechanics-informed variational autoencoder (MI-VAE), which integrates biomechanical parameters from esophageal diagnostics like FLIP [27]. Their approach mapped esophageal motility disorders into a Virtual Disease Landscape (VDL) that accurately clustered patients by condition and predicted treatment outcomes, such as the effectiveness of peroral endoscopic myotomy (POEM) or pneumatic dilation, based on esophageal wall stiffness and contraction strength. For achalasia patients, the model quantitatively assessed the reduction in esophageal sphincter tone and improvement in contraction patterns after POEM, showing a significant post-treatment shift towards a normal motility profile. Additionally, the MI-VAE framework enabled longitudinal tracking of patients' condition over time, highlighting its potential for optimizing personalized treatment strategies and monitoring long-term outcomes.

3.2 | Esophagogastric Junction Outflow Obstruction

A diagnosis of EGJOO may be given after observing a specific manometric pattern (elevated median integrated relaxation pressure with intact or weak peristalsis) and is often misdiagnosed due to overlapping features with other disorders such as achalasia [13, 28]. AI has significantly improved its diagnostic precision by addressing these ambiguities [13].

AI may be applied in settings where diagnoses need to be made more confidently, such as in the case of differentiating EGJOO from benign causes of esophagogastric junction metrics. It may also help with distinguishing which patients will have symptom resolution and who will progress [13]. Leveraging CNNs has shown potential in automating key diagnostic metrics with high accuracy, which is crucial for conditions like EGJOO [18]. These advancements may help adjust diagnostic procedures for EGJOO and improve patient outcomes.

3.3 | Ineffective Esophageal Motility

IEM is characterized by weak or absent peristalsis, making it one of the most challenging motility disorders to diagnose due to its subtle manifestations [29]. Despite being one of the most common findings in manometry, IEM often presents diagnostic ambiguities, particularly in distinguishing clinically significant cases from incidental findings [30]. It is characterized by diminished esophageal peristaltic vigor and multiple weak, absent, or fragmented peristalsis on HREM [30]. AI has improved the diagnostic process by enabling precise analysis of weak peristalsis patterns. In a study by Santos et al. [31] evaluating the feasibility of artificial neural networks (ANN) for esophageal motility disorders, IEM was reliably classified, with diagnostic accuracies reaching 72%–82% depending on validation techniques. The ANN-based models leveraged advanced mathematical relationships within manometric data, enabling the identification of subtle abnormalities in distal esophageal contractions, often overlooked during manual interpretations. By reducing inter- and intra-observer variability, ANNs addressed a significant limitation in traditional diagnostic methods, improving consistency and reliability across classifications. These findings suggest that AI tools, such as ANNs, have the potential to transform IEM diagnosis by automating classification, enhancing sensitivity, and enabling more accurate differentiation of clinically significant cases from incidental findings. Addressing interobserver variability through AI is of distinct importance in this diagnosis, especially due to its elusive nature and its ability to mimic other disorders [11, 22].

The enhanced ability to detect and classify IEM has critical implications for treatment planning. Patients with mild IEM may benefit from dietary modifications and prokinetics, while severe cases can be linked to systemic conditions like scleroderma, requiring targeted therapy. By identifying these distinctions, AI tools are transforming patient outcomes through these domains.

3.4 | Distal Esophageal Spasm

DES involves uncoordinated or premature contractions of the distal esophagus, resulting in chest pain or dysphagia. In manometry, it is characterized by simultaneous contractions in the distal esophagus in 20% or more swallows, alternating with normal peristaltic events [32]. AI has been particularly effective in addressing the challenges of diagnosing DES, which often presents episodically and may be overlooked in conventional analysis. Popa et al. discussed a CNN that achieved over 90% accuracy in identifying simultaneous and uncoordinated contractions, crucial diagnostic features of DES [18]. The model's specificity in distinguishing DES from similar motility disorders, such as achalasia, underscores its precision in refining differential diagnoses. Kou's research introduced long short-term memory (LSTM) networks capable of analyzing large datasets and providing real-time visual overlays on HREM topography, enabling immediate recognition of simultaneous and fragmented contractions characteristic of DES [15]. This real-time feedback

greatly enhances diagnostic efficiency and reduces the cognitive burden on clinicians. Surdea-Blaga et al. discussed the ability of AI to reduce false-positive DES diagnoses by standardizing contraction vigor and propagation speed thresholds, minimizing interobserver variability and increasing diagnostic consistency [14]. Furthermore, AI-driven models have refined distal latency metrics, enhancing the detection of spastic activity by offering more precise temporal measurements, as emphasized by Surdea-Blaga et al. Additionally, Popa and Kou demonstrated the integration of pressure and impedance metrics using AI, providing a comprehensive understanding of bolus transit and contraction vigor, critical for distinguishing DES from other spastic motility disorders [15, 18]. Kou's LSTM models further advanced visual mapping capabilities, enabling visualization of spastic contractions, which enhances the clinician's ability to identify subtle motility abnormalities [15]. Together, these advancements illustrate the transformative potential of AI in accurately diagnosing and managing DES by addressing key diagnostic challenges.

3.5 | Jackhammer Esophagus and High-Pressure Disorders

Jackhammer esophagus, a hypercontractile motility disorder, is defined as at least two swallows with DCI > 8000 mmHg·s·cm [23]. AI systems analyzing contractile vigor can improve the classification and management of this condition, especially due to its overlap with other diseases. ML algorithms that apply to many other motility disorders may also be applied in this context. As mentioned by Zifan et al. AI has demonstrated high accuracy in calculating DCI and analyzing hypercontractile patterns in HREM with precision and consistency [21]. Quantifying hypercontractile activity by AI is therefore possible and is a critical metric in diagnosing this high-pressure disorder.

4 | Automated Assessment of Impedance-pH Tracings

AI offers transformative potential in the evaluation of impedance-pH tracings, as evidenced by its ability to automate and enhance diagnostic workflows in GERD. A study by Rogers et al. highlighted the effectiveness of an AI-based decision tree algorithm in identifying and censoring esophageal events, such as swallows, belches, and reflux episodes, to produce artifact-free BI values [19]. This capability allowed the calculation of upright and recumbent BI, revealing that the upright-to-recumbent BI ratio (U:R AIBI ratio) correlated strongly with GERD treatment outcomes. Results demonstrated that a U:R AIBI ratio ≥ 1.0 was a significant predictor of symptom improvement, outperforming traditional metrics like mean nocturnal baseline impedance (MNBI) in specific subsets of patients, particularly those with borderline acid exposure time (AET) [19]. Moreover, the AI algorithm provided instantaneous, reproducible metrics, improving diagnostic accuracy by mitigating the confounding effects of retained esophageal content during supine measurements [19]. The strong correlation between recumbent AI-derived BI and MNBI validated the algorithm's accuracy, while the distinct diagnostic value of upright BI emphasized the utility of AI in uncovering novel metrics for GERD evaluation. These findings suggest that AI not only optimizes traditional pH-impedance analysis but also introduces innovative approaches to understanding GERD pathophysiology and predicting therapeutic outcomes.

5 | Functional Dysphagia

Per the ROME criteria, functional dysphagia is defined as a sensation of abnormal bolus transit with no evidence of structural lesions or major motor disorders [33]. AI has emerged as a transformative tool in the assessment and management of functional dysphagia, offering innovative solutions to longstanding challenges in the field. ML techniques have demonstrated the potential to enhance diagnostic precision and enable non-invasive, continuous monitoring of swallowing activities. For instance, Zifan et al. developed swallowing activity classifiers using acoustic recordings analyzed by ML algorithms such as multi-layer perceptrons, CNN, and Convolutional Recurrent Neural Networks (CRNN). These models effectively classify swallowing-related activities like choking, coughing, and swallowing, achieving validation accuracies of up to 74%. This underscores their potential applicability in clinical settings and remote monitoring for elderly populations with swallowing difficulties [21].

Further advancements integrate AI into the analysis of high-resolution impedance planimetry and manometry data, as described in studies on the FLIP. AI frameworks utilize clinical and mechanics-based metrics, such as contraction power and esophagogastric junction obstruction probabilities, to provide unified diagnostic insights [27]. These tools classify functional and spastic-reactive motility disorders with enhanced precision, utilizing probabilistic modeling and virtual disease landscapes to refine diagnosis [27].

AI also enhances biofeedback systems for dysphagia therapy, where it improves the detection of swallowing onsets through combined electromyography (EMG) and bioimpedance (BI) data. As demonstrated in recent studies, AI-based systems outperform traditional threshold-based methods, achieving higher F1 scores and reducing false-positive rates, thereby improving real-time applications in biofeedback and functional electrical stimulation (FES). These advancements hold promise for both clinical and at-home care applications, addressing the growing demand for personalized and efficient dysphagia interventions [27, 34].

6 | Predictive Models and Therapeutic Outcomes

While no direct studies currently examine AI's role in predicting therapeutic outcomes in esophageal motility disorders, insights can be extrapolated from other applications in medicine. AI's ability to integrate large datasets and analyze complex variables positions it as a promising tool for predicting patient outcomes and optimizing therapies. For instance, AI has been applied successfully in acute ischemic stroke management to predict therapeutic outcomes, such as response to endovascular thrombectomy, by integrating imaging and clinical data to assist decision-making [35]. Similarly, in the context of orthognathic surgery, AI demonstrated efficacy in

predicting post-surgical soft tissue outcomes, leveraging CNNs to improve accuracy and patient-specific predictions [36]. These successes highlight AI's potential to model therapeutic outcomes by integrating clinical, imaging, and patient-specific factors in complex disorders and give hope to its applications in motility disorders as well.

Moreover, AI's applications in prognostication for stroke outcomes illustrate its capability to process heterogeneous data streams and predict long-term functional outcomes, which could be extrapolated to therapeutic response modeling in esophageal motility disorders [35]. The adaptability of AI to predict responses to interventions, as seen in dental and craniofacial applications [36], may support its utility in tailoring treatment plans and forecasting efficacy in gastrointestinal disorders.

Finally, the demonstrated utility of AI in various medical fields underscores the importance of developing robust datasets and incorporating patient-specific variables to optimize predictive models [37]. By extrapolation, AI could revolutionize esophageal motility disorders' therapeutic outcomes by aiding in personalized treatment strategies and improving clinical decision-making processes.

7 | Educational Applications of AI in Esophageal Testing

AI is not only a diagnostic tool but also a valuable resource in medical education, particularly for training in esophageal motility interpretation. The complexity of HREM, reflux monitoring, and related technologies often makes proficiency challenging for trainees.

AI-enhanced training tools have been applied in other areas of medical education and could be extrapolated to motility disorders. For instance, an AI-driven radiology simulator demonstrated a 35% improvement in trainees' diagnostic accuracy and a reduction in interpretation time by 25% after just five sessions, underscoring AI's capability to enhance learning efficiency and confidence [38].

In gastroenterology, AI-powered platforms have successfully facilitated virtual endoscopy training, enabling learners to achieve a 20% faster procedural competence with comparable outcomes to traditional hands-on methods [39]. Furthermore, in histopathology education, AI systems have been shown to improve trainees' ability to identify complex patterns, such as distinguishing rare cancer subtypes, with accuracy rates increasing by 18% compared to manual methods [40].

Drawing from these results on esophageal motility training, AI could be employed to identify motility patterns, simulate clinical scenarios, and offer feedback on diagnostic decisions. Such systems could reduce the learning curve associated with HREM, ensure standardization in training, and ultimately improve diagnostic proficiency in esophageal motility disorders.

Lastly, AI-powered virtual mentorship could theoretically provide support to trainees performing HREM studies while they are performing them, ensuring accuracy even in the absence of direct supervision. Such innovations hold the potential to expand access to high-quality training globally.

8 | Conclusion

Artificial intelligence has emerged as a transformative force in the diagnosis, management, and research of esophageal motility disorders. By enhancing diagnostic precision, standardizing workflows, and enabling predictive analytics, AI addresses many longstanding challenges in this field. AI is changing the game in diagnosing, managing, and researching esophageal motility disorders. It is helping to improve accuracy, streamline workflows, and predict outcomes, addressing long-standing challenges in the field. That said, there is still room for growth, especially when it comes to refining algorithms, testing them in different patient groups, and combining various data types. Continued innovation in AI applications, coupled with multicenter validations and broader accessibility, as well as more widespread access and real-world validation, will ensure its integration into routine clinical practice, positioning AI as a key player in everyday healthcare. In doing so, it will help doctors deliver more personalized care and ultimately improve patient outcomes.

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The authors have nothing to report.

Consent

The authors have nothing to report.

Conflicts of Interest

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The authors have nothing to report.

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